Group Details

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Github Repository URL

https://github.com/yuvraj56j-del/Digital-wallet-payment-and-consumer-choice-project.git

Background and problem

Digital wallets, USSD and QR payments are now a routine way to pay in Kenya, so wallet providers, banks and merchants need to know what customers buy, how they pay and where payments break down. Failed, pending and reversed transactions cost users time and money and cost providers trust.

This project analyses a year of synthetic wallet transactions (2025) to describe spending patterns, usage trends and payment failures. The intended users are a wallet provider's product team, a bank or fintech working to reduce failed payments, and merchants who want to know when and how customers pay.

Digital Wallet Transactions and Consumer Habits – Project 6

Target Users

The findings will be for people who run or build digital payment services in Kenya who include:

- A digital wallet provider’s product team.
- A bank or a fintech trying to cut failed payments.
- Merchants and business owners who want to know when and how customers pay for their products.

Overall Objective of our project

To analyse 2025 digital wallet transactions in Kenya(using synthetic data) in order to understand what customers spend on, how, where and when they pay, and where payments fail. The aim is to recommend ways to improve the experience of using the digital wallet among customers.

The 3 Specific Objectives

1. Identify the spending categories, customer segments and regions that account for most transactional value.
2. Describe how transaction volume and value change over time (month, weekday, hour)
3. Compare payment channels and regions by failure, pending and the reversal rates.

Research Questions (all are tied to one of the 3 specific objectives above)

1. Which merchant categories account for the largest share of total spending in Ksh? – Objective 1
2. Which customer segments have the highest median transaction value? – Objective 1
3. Which weekday and which hour of the day are the busiest? – Objective 2
4. Which payment channels have the highest failure rate? – Objective 3
5. Which regions record the most transactions, and which have the highest average value? – Objective 1
6. How do total monthly transaction value and transaction value and transaction count change across 2025? – Objective 2
7. Which regions have the highest success/failure rate? – Objective 3